DATA 298A: End-to-End Data Pipeline Demonstration

Personalized Education Agent – Team 2

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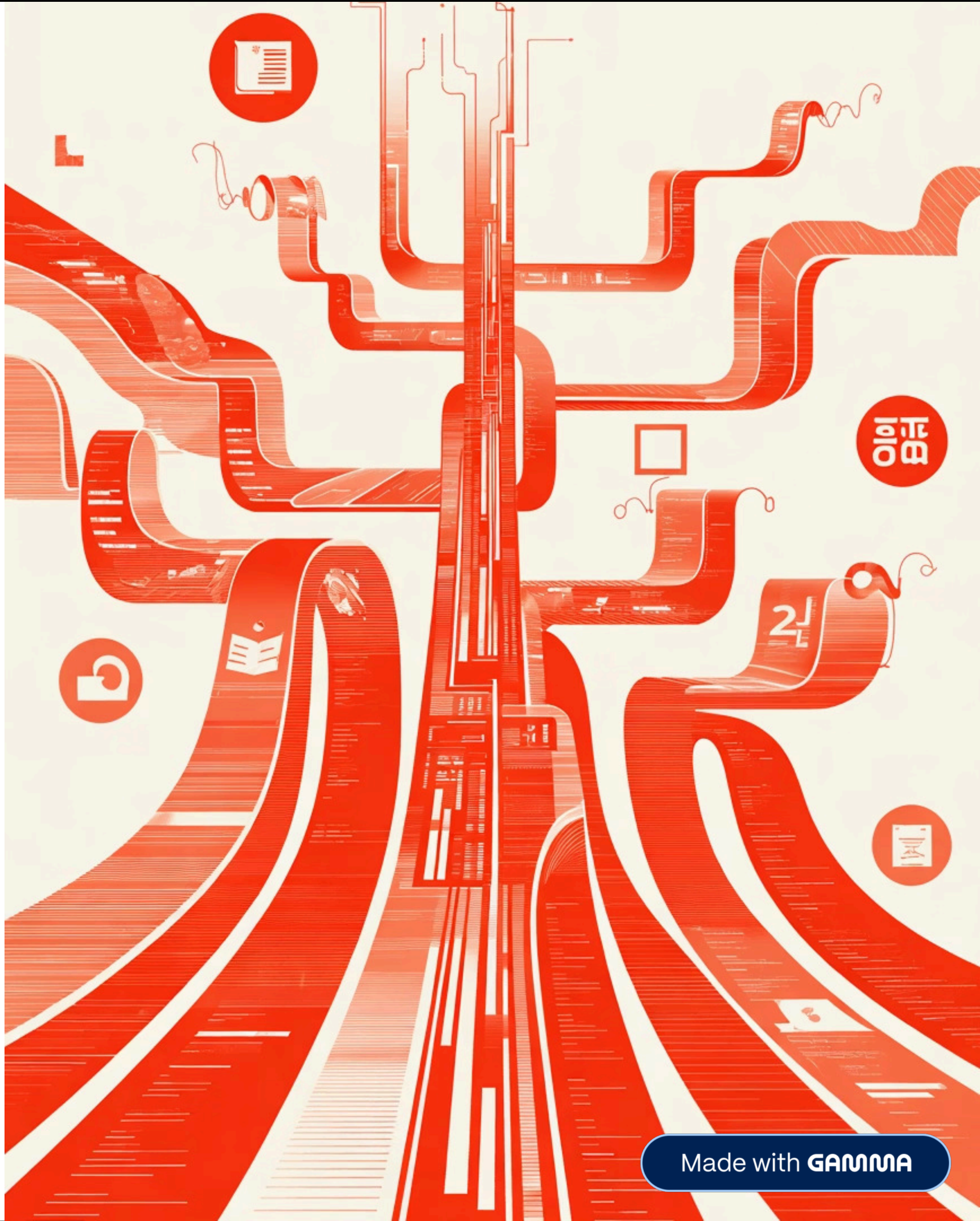


3.1 Data Process

Six-Stage Data Pipeline Architecture

End-to-End Data Processing Pipeline (from raw → prepared)

01

Source Identification & Access

Identifying and accessing relevant data sources for the personalized education agent

02

Ingestion & Snapshotting

Capturing and storing raw data with version control and timestamps

03

Cleaning & Validation

Removing inconsistencies, duplicates, and validating data quality

04

Transformation & Feature Engineering

Converting raw data into meaningful features for model training

05

Dataset Preparation & Splitting

Creating train, validation, and test sets with proper partitioning

06

Storage, Versioning & Documentation

Maintaining data lineage and reproducibility with DVC and MLflow

Note: Workflow tracked with DVC + MLflow ensuring traceability and reproducibility.

3.2 Data Collection

Data Collection Summary (Diversity & Scale)

Dataset	Rows	Size	Description
LeetCode	10k	80MB	Programming problems
APPS	9k	75MB	Academic code tasks
CodeSearchNet	12k	90MB	Code-text pairs
CodeForces	8k	65MB	Competitive programming logs

9M+ records

~11 GB raw data collected

98.6% KC coverage

99% schema validation

3.2 Data Collection

Representative Raw Data Sample ("Before")

Mixed Schemas

Inconsistent field names and data types across different sources before standardization

Nulls & Inconsistencies

Missing values and text inconsistencies requiring cleaning and validation

Ethical Compliance

PII scrubbed; all user IDs hashed to ensure privacy and data protection

Highlight: Mixed schemas, nulls, and text inconsistencies before cleaning. Ethical compliance: PII scrubbed; all user IDs hashed.

3.3 Pre-processing

Pre-processing Workflow & Tools

Raw

Unprocessed data from multiple sources

Structured

Organized into consistent schemas

Cleaned

Validated and ready for transformation

Key Tools

- Pandas
- Spark
- Great Expectations

Results Summary

- **99.5%** Schema Pass Rate
- **<0.5%** Duplicates Removed
- **100%** KC Mapping Coverage

Visual: "Data Cleaning & Validation Flowchart" (Fig. 3.3.7).

3.3 Pre-processing

Representative Cleaned Dataset

problem_id	title	difficulty	tags	kc_tags	runtime
LC001	Two Sum	Easy	arrays, hash-map	data-structures	1200ms
LC042	Trapping Rain Water	Hard	arrays, two-pointers	algorithms	2400ms
CF156	Dynamic Programming	Medium	dp, optimization	problem-solving	1800ms

Unified Schema

Consistent field names and data types across all datasets

Normalized Timestamps

ISO 8601 UTC format for temporal consistency

Model-Ready Fields

Clean, validated data prepared for machine learning

Before/After callout: "Raw data transformed into reliable, uniform input for modeling."

3.4 Transformation

Transformation & Feature Construction

Example – Before vs After

Field	Before	After
difficulty	"medium"	1
tags	["arrays", "hash-map"]	"arrays, hash-map"
runtime	"1.2 s"	1200
timestamp	"15/10/25 3:05PM"	ISO 8601 UTC

Tools

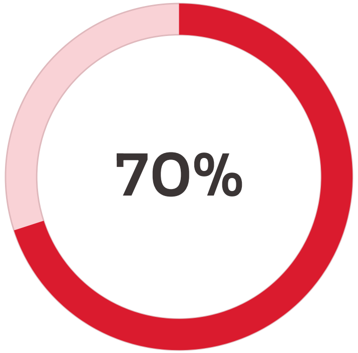
- Spark
- MLflow
- Python scripts (transform_sequences.py)

Outcome

9.3M records transformed, tracked, validated, versioned

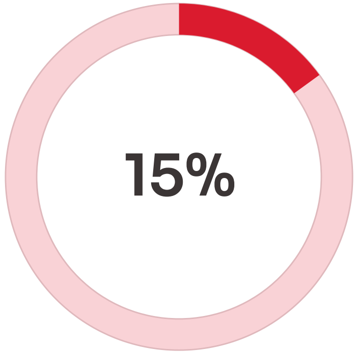
3.5 Data Preparation

Train / Validation / Test Construction



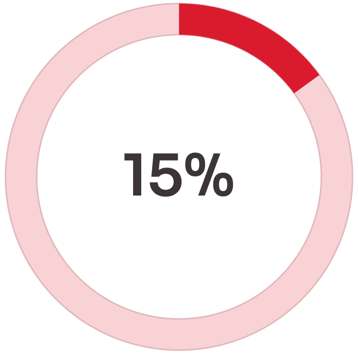
Training Set

Primary dataset for model learning



Validation Set

Hyperparameter tuning and model selection



Test Set

Final performance evaluation

Leakage Prevention

Risk	Mitigation
Same user across splits	User-level partitioning
Temporal leakage	Chronological split
Duplicate questions	Hash-based filtering

Metrics: 99.8% schema pass, 0.21% duplicates, 0.05% missing.

3.6 Data Statistics

Data Statistics Overview

Summary Table

Stage	Total Records	Key Change
Raw	9.3M	Initial ingestion
Cleaned	9.25M	47,832 duplicates removed
Transformed	7.1M KT + 12k QA	Added features
Prepared	9.3M	Train/Val/Test split

Visuals: Distribution plot (difficulty levels, KCs, languages).

Note: "High-quality, balanced, model-ready datasets."

End-to-End Pipeline Complete

From 9.3M Raw Records to Model-Ready Data

9.3M	99.8%	100%	6
Total Records	**Schema Pass Rate**	**KC Coverage**	**Pipeline Stages**
Processed and prepared	High-quality validation	Complete knowledge component mapping	Comprehensive data processing

Outcome: Traceable, reproducible, and production-ready data pipeline with DVC + MLflow integration ensuring full lineage and versioning capabilities.